

For temperature

We can calculate the entropy for the attribute "Temperature" with values Hot, mild, cool.

For Hot 2 "Yes" 2 "No".

$$\text{Entropy (Hot)} = \left(-\frac{2}{4}\right) \log_2\left(\frac{2}{4}\right) + \left(-\frac{2}{4}\right) \log_2\left(\frac{2}{4}\right) \\ = 1$$

For mild 4 "Yes" 2 "No"

$$\text{Entropy (mild)} = \left(-\frac{4}{6}\right) \log_2\left(\frac{4}{6}\right) + \left(-\frac{2}{6}\right) \log_2\left(\frac{2}{6}\right)$$

$$\approx 0.918$$

For cool

3 "Yes" 1 "No"

$$\text{Entropy (cool)} = \left(-\frac{3}{4}\right) \log_2\left(\frac{3}{4}\right) + \left(-\frac{1}{4}\right) \log_2\left(\frac{1}{4}\right)$$

$$\approx 0.811$$

Weighted average entropy for "Temperature"

$$\text{Entropy (Temperature)} = \frac{4}{14} \times 1 + \frac{6}{14} \times 0.918 + \frac{4}{14} \times 0.811$$

$$\approx 0.911$$

The Information Gain for temperature

$$\text{Gain (S, temperature)} = 0.940 - 0.911$$

$$= 0.029$$

3. Choosing the best attribute to split

$$\text{Outlook} = 0.249$$

$$\text{Humidity} = 0.152$$

$$\text{Windy} = 0.048$$

$$\text{Temperature} = 0.029$$

Since Outlook has highest information gain (0.249) we choose Outlook as the root for the decision tree

4. Split on Outlook

Now, we split the dataset based on "outlook". There are three possible branches. Sunny, Overcast and Rain

- Overcast: All outcomes are "Yes". So it becomes leaf node with the label "Yes".

- Sunny: Further split the data for the "Sunny" branch

- Rain: Further split the data for the "Rain" branch

5. Split on outlook Sunny

For the Sunny branch remaining data is:

Outlook	Temperature	Humidity	Windy	Play Golf
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Sunny	Mild	Normal	True	Yes

For hot

Target value is always "No". Since this subset is pure
entropy is 0

$$\text{Entropy (Hot)} = 0.0$$

For mild

One outcome is Yes and one outcome is No

$$\begin{aligned}\text{Entropy (mild)} &= \left(-\frac{1}{2}\right) \log_2\left(\frac{1}{2}\right) + \left(-\frac{1}{2}\right) \log_2\left(\frac{1}{2}\right) \\ &= 1.0\end{aligned}$$

For cool

target is Yes, $\therefore$ Entropy is zero

$$\text{Entropy (cool)} = 0.0$$

Information Gain is

$$\text{Gain}(S_{\text{sunny}}, \text{Temp}) = \text{Entropy}(S_{\text{sunny}}) - \sum_{v \in \{\text{Hot, mild, cool}\}} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$

$$\begin{aligned}\text{Gain}(S_{\text{sunny}}, \text{Temp}) &= 0.97 - \left(\frac{2}{5} \times \text{Entropy}(S_{\text{Hot}}) + \frac{2}{5} \times \text{Entropy}(S_{\text{mild}})\right. \\ &\quad \left.+ \frac{1}{5} \times \text{Entropy}(S_{\text{cool}})\right)\end{aligned}$$

$$= 0.97 - \left(\frac{2}{5} \times 0.0 + \frac{2}{5} \times 1.0 + \frac{1}{5} \times 0.0\right)$$

$$= 0.97 - 0.4$$

$$= 0.57$$

For Humidity

High humidity: All No, so it becomes a leaf node

Normal humidity: All Yes, so it becomes a leaf node

$$\text{Entropy (High)} = 0.0$$

$$\text{Entropy (Normal)} = 0.0$$

$$\text{Gain} = 0.97 - 0$$

$$\left(\frac{1}{5} \log_2 \left(\frac{1}{5} \right) \right) + \left(\frac{4}{5} \log_2 \left(\frac{4}{5} \right) \right) = 0.97$$

For Wind

Strong: 1 "Yes" 1 "No"

$$\text{Entropy (Strong)} = 1$$

$$\text{Entropy (Weak)} = \left(\frac{1}{3} \right) \log_2 \left(\frac{1}{3} \right) + \left(\frac{2}{3} \right) \log_2 \left(\frac{2}{3} \right)$$

$$= 0.9183$$

$$\text{Gain (Sunny, Wind)} = 0.97 - \frac{2}{5} \times 1.0 - \frac{3}{5} \times 0.9183$$

$$= 0.0192$$

$$\left(0.0 \times \frac{1}{5} + 0.1 \times \frac{2}{5} + 0.0 \times \frac{3}{5} \right) = 0.0$$

$$0.0 - 0.0$$

$$= 0.0$$